Computer Organization & Design

Lecture / Lab-006

Logical Operations Using Integrated Circuits on Breadboard

Learning Objectives

- The purpose of this lab is to understand the behavior of basic logic gates
- Implementing **NOT, AND, and OR gates** using digital logic ICs on a breadboard.
- Students will verify the truth tables by observing LED outputs.

Materials

- Breadboard
- Jumper wires
- 5V power supply
- LEDs
- 330Ω resistors
- Breadboard power supply
- Logic ICs:











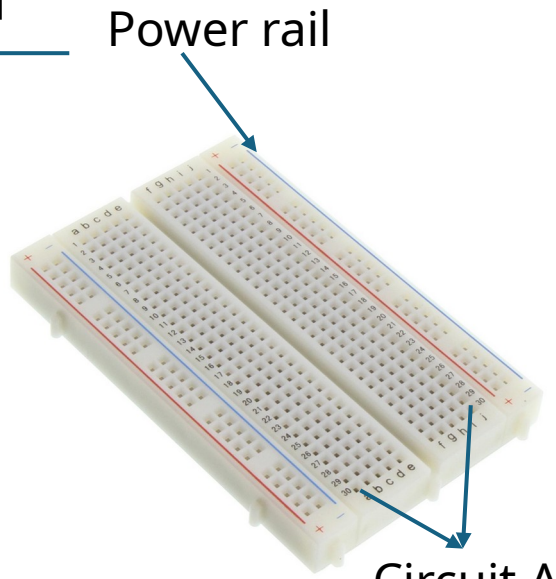


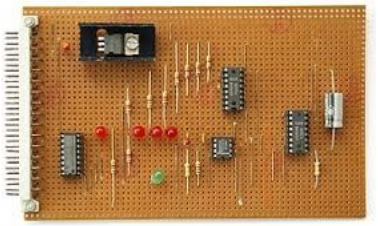
Gate	IC
NOT	**7404**
AND	**7408**
OR	**7432**

Component Description

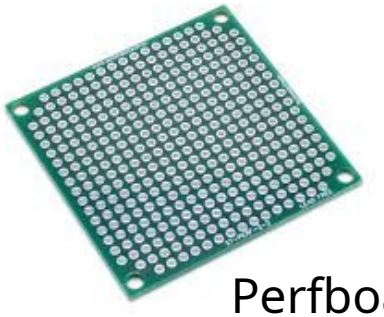
Breadboard

- Used to build and test circuits without soldering





Components soldered on perfboard



Perfboard

Power rails

- Red line → VCC (+ voltage)
- Blue line → GND

Terminal strips (middle area)

- The center section is where components go.
- Each row of 5 holes is connected horizontally:
- The center gap separates the two halves so you can place IC chips.

Center gap

- Used for ICs like logic chips.
- Pins on opposite sides are not connected, which prevents short circuits.

Board	Purpose	Connection method
Breadboard	Temporary testing	No soldering
Perfboard / Proto board	Permanent circuit	Solder components
PCB (Printed Circuit	Professional design	Manufactured copper





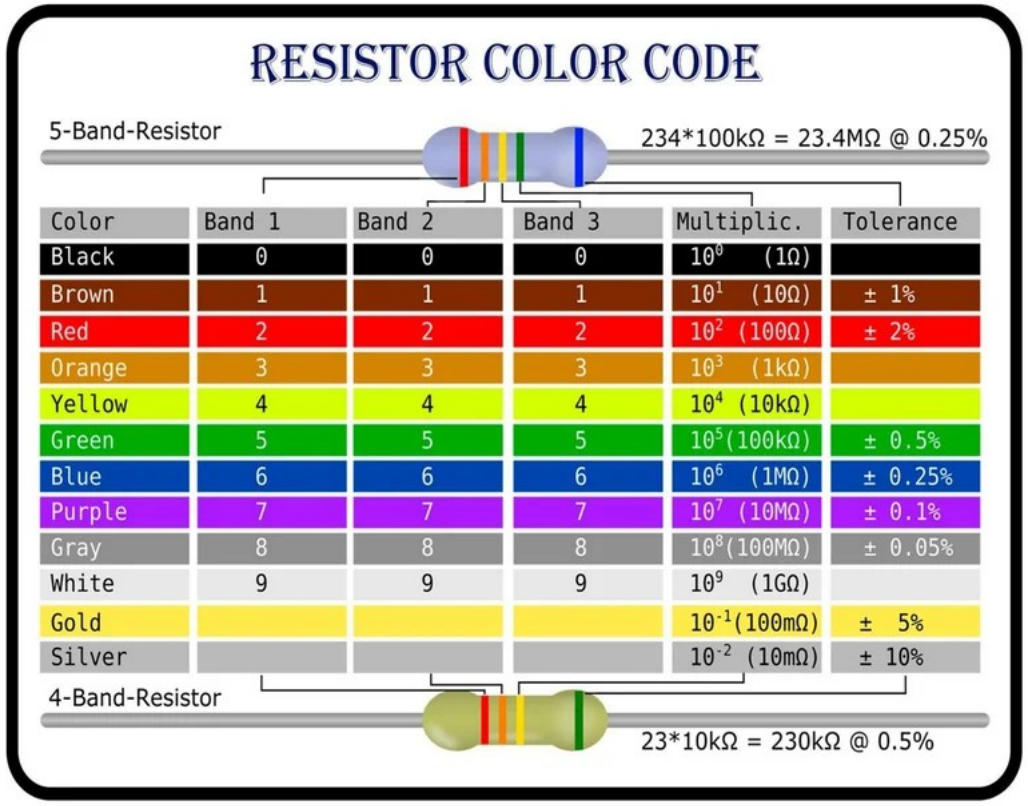
PCB fabrication

- Copper coating
- Photolithography
- Chemical Etching

Component Description



- Resistor color code



Color	Band 1	Band 2	Band 3	Multiplic.	Tolerance
Black	0	0	0	10^0 (1Ω)	
Brown	1	1	1	10^1 (10Ω)	± 1%
Red	2	2	2	10^2 (100Ω)	± 2%
Orange	3	3	3	10^3 (1kΩ)	
Yellow	4	4	4	10^4 (10kΩ)	
Green	5	5	5	10^5(100kΩ)	± 0.5%
Blue	6	6	6	10^6 (1MΩ)	± 0.25%
Purple	7	7	7	10^7 (10MΩ)	± 0.1%
Gray	8	8	8	10^8(100MΩ)	± 0.05%
White	9	9	9	10^9 (1GΩ)	
Gold				10^{-1}(100mΩ)	± 5%
Silver				10^{-2} (10mΩ)	± 10%

Order (left → right):

- 4-Band Resistor
  - $Digit_1$ – $Digit_2$ – Multiplier – Tolerance
- 5-Band Resistor
  - $Digit_1$ – $Digit_2$ – $Digit_3$ – Multiplier – Tolerance

Example:

- Red – Violet – Brown – Gold
  - 27 ×10 = 270 ohms(±5%)
- Orange – Orange – Brown – Gold
  - 33 ×10 = 330 ohms(±5%)
- Brown – Black – Black – Red – Brown
  - 100 × 100 = 10,000ohms= 10kohms(±1%)

- What is readings of resistor with color code Red – Black – Red – Gold?

20 × 100 = 2,000ohms= 2kohms(±5%)

Component Description

- LED (light emitting diode)

Terminal	Name	Function
Anode (+)	Positive	Connects to Vcc through a resistor
Cathode (−)	Negative	Connects to GND

How to identify them

- Long leg → Anode (+)
- Short leg → Cathode (−)

Current flows:

Vcc → resistor → LED (anode → cathode) → GND



- 9v Battery



Terminal	Symbol	Role
Positive	+	Supplies voltage
Negative	–	Return path (ground)

Example: Lighting an LED with a 9V battery

- Use a resistor to limit current.
  - +9V → 330Ω resistor → LED (long leg) → LED short leg → battery
- Recommended resistor:
  - 330 Ω – 1 kΩ
- Without a resistor, the LED will draw too much current and burn out immediately.

Component Description

Breadboard power supply module (MB102)

Part	Function
Barrel jack	Connect a 9V–12V DC adapter
USB port	Alternative power input
Voltage regulators	Output 5V or 3.3V
Jumpers (yellow pins)	Select 5V or 3.3V for each rail
Power switch	Turn power on/off
LED	Shows power is on

- Input: 9V battery or 9–12V adapter
- Output:
  - 5V (for most logic chips like 74LS)
  - 3.3V (for some sensors and microcontrollers)

A 9V battery can power it, but it drains quickly.
A 9V or 12V wall adapter works much better.



- Jumper wires
  - Connect components on a breadboard or between modules.

Type	Use
Male → Male	Breadboard to breadboard
Male → Female	Breadboard to module pins
Female → Female	Module to module

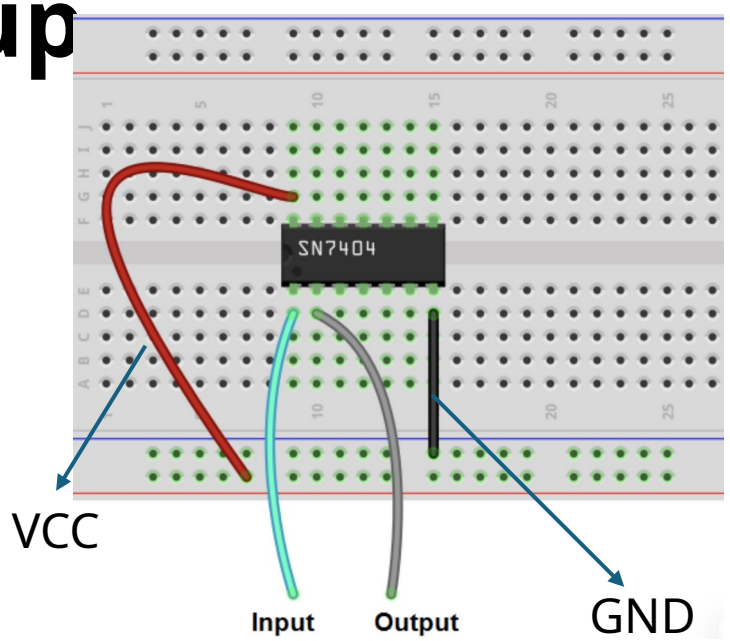


Part 1: Breadboard Power Setup

- Connect 5V to the breadboard positive rail.
- Connect GND to the ground rail.
- Place the IC across the center gap of the breadboard.

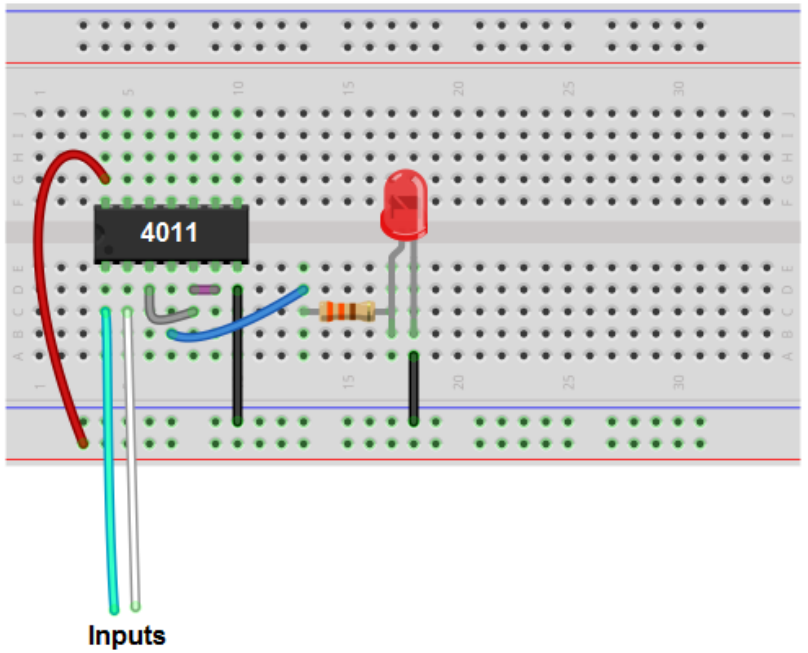
For all **74xx ICs**:

- Pin 14 → Vcc (5V)
- Pin 7 → GND



- VCC = where voltage comes from (+5V)->Power supply
- GND = where curent returns (0V reference)->Ground

Current moves from VCC-> components->GND



Part 2: NOT Gate (7404)

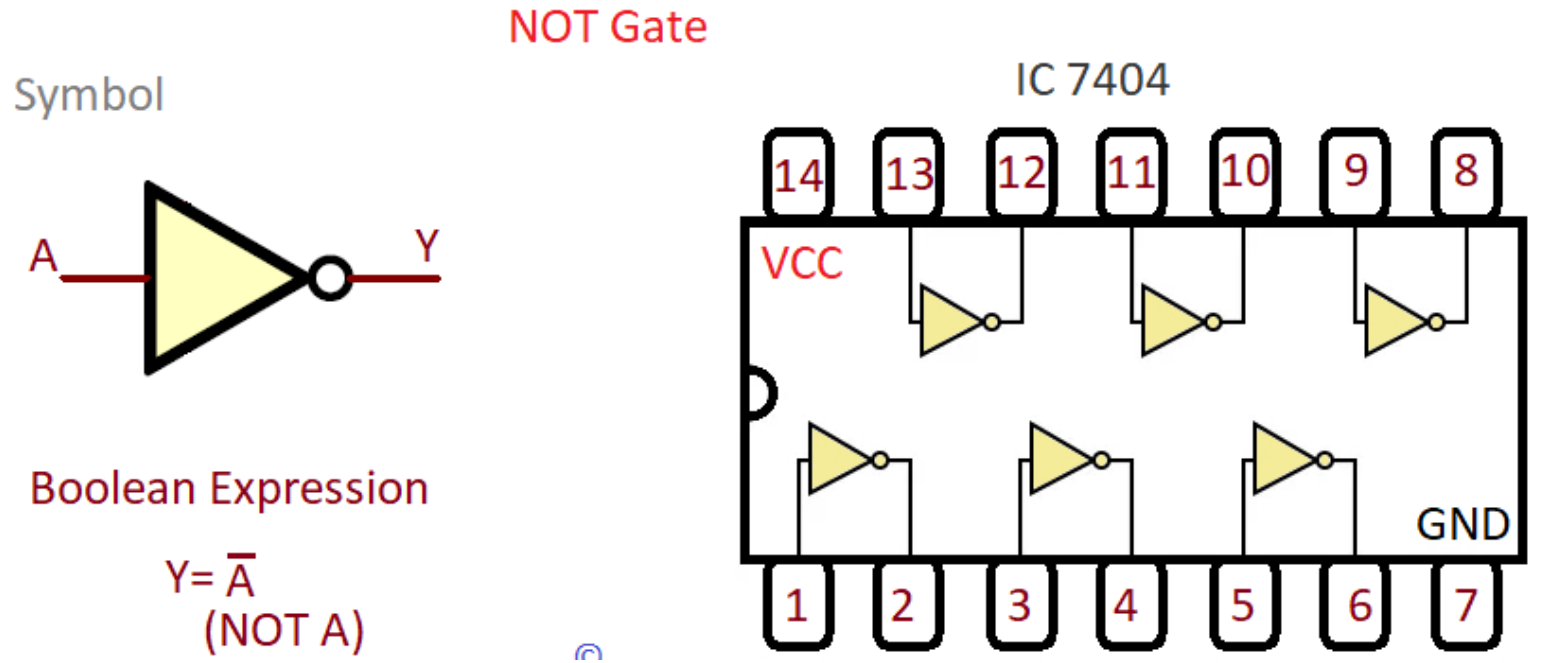
- **Gate behavior**

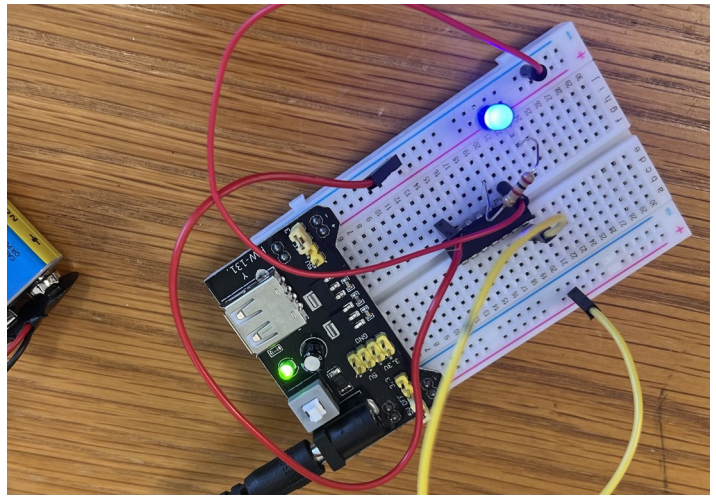
	A	NOT A
VCC	0	1 LED GLOWS
GND	1	0 OFF

- **Example pins**
  Pin 1 → input A
  Pin 2 → output NOT A

- **Steps**
  Connect pin 14 → 5V
  Connect pin 7 → GND
  Connect pin 1 → input switch
  Connect pin 2 → LED → resistor → GND
  Toggle the input and observe the output LED.

- **Expected behavior:**
  Input LOW → LED ON
  Input HIGH → LED OFF





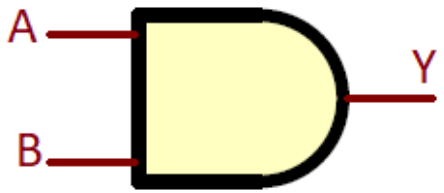
Part 3: AND Gate (7408)

- **Truth Table**

A	B	A AND B
0	0	0
0	1	0
1	0	0
1	1	1

Logic AND Gate

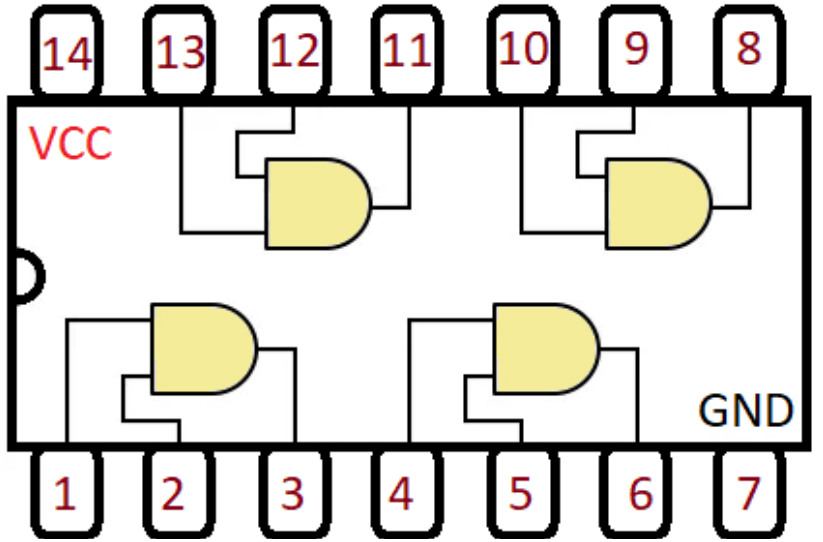
Symbol



Boolean Expression

Y= A.B

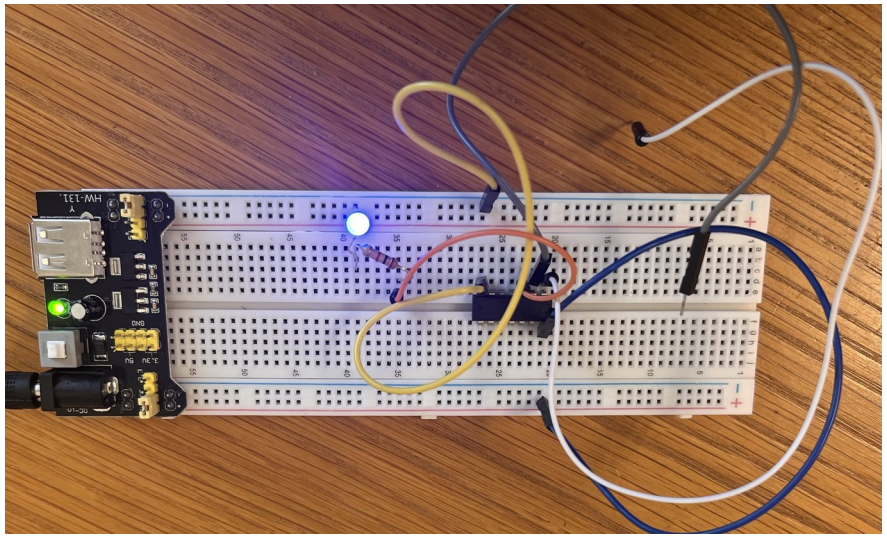
IC 7408



- **Gate pins**
  Pin 1 → input A
  Pin 2 → input B
  Pin 3 → output

- **Steps**
  Connect two input wires or switches to pins 1 and 2.
  Connect pin 3 → LED → resistor → GND.
  Test all input combinations.

  Only when A=1 and B=1 should the LED turn on.



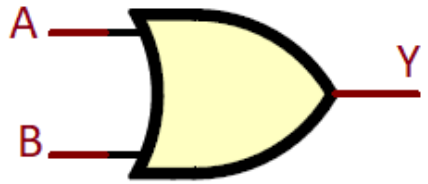
Part 4: OR Gate (7432)

- **Truth Table**

A	B	A OR B
0	0	0
0	1	1
1	0	1
1	1	1

Logic OR Gate

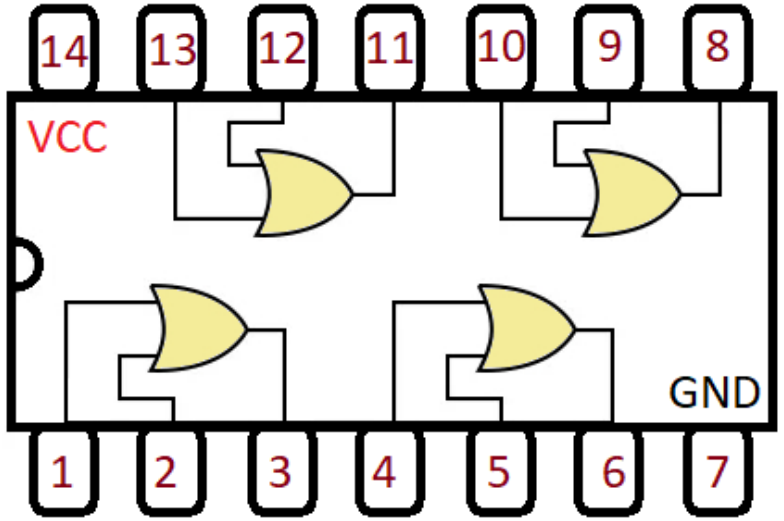
Symbol



Boolean Expression

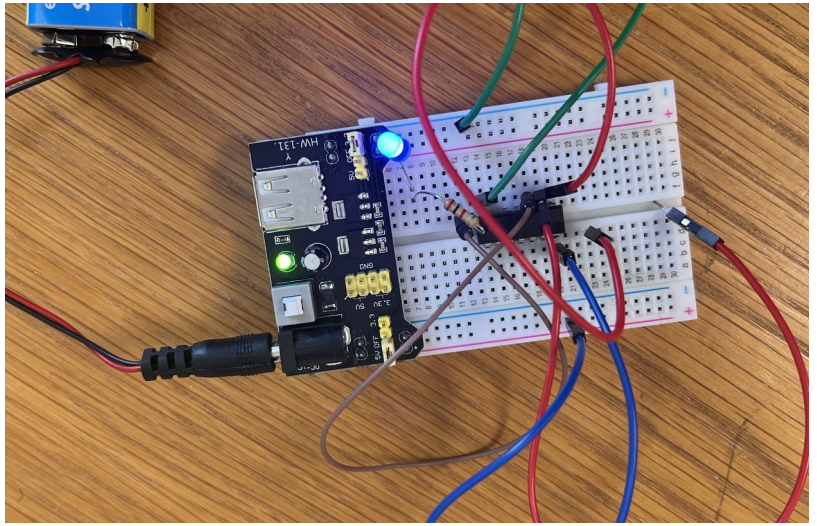
Y= A+B

IC 7432



©

- **Gate pins**
  Pin 1 → input A
  Pin 2 → input B
  Pin 3 → output



- **Steps**
  Same setup as the AND gate but using the 7432 IC.

  LED should turn on if either input is HIGH.

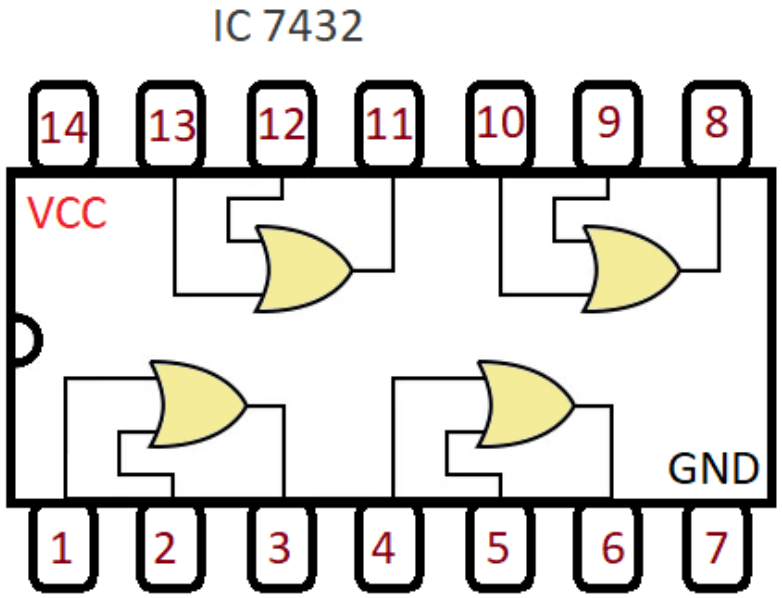
Hardware Gates vs. MIPS Instructions

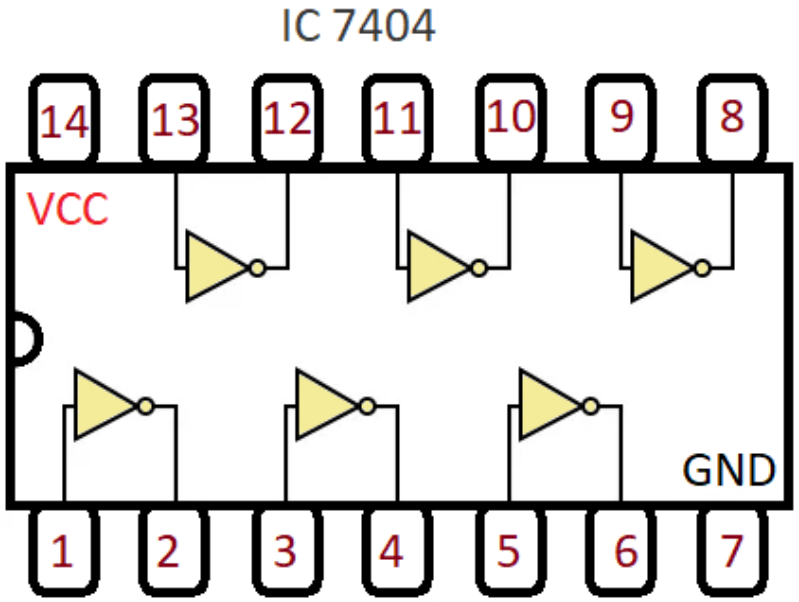
Gate	MIPS instruction
AND	and
OR	or
NOT	nor with $zero

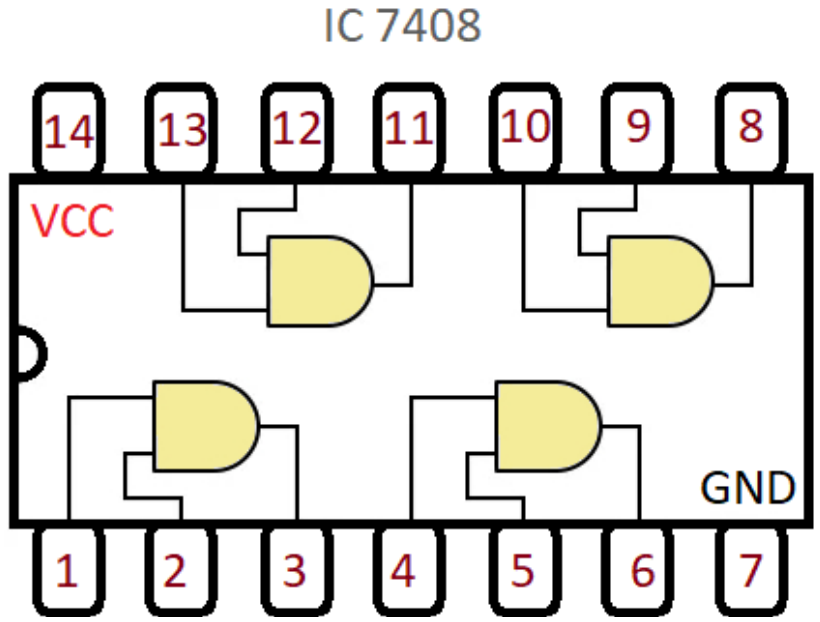
- Example:
  nor $t0, $t1, $zero # bitwise NOT

In-class Task/Assignment

1. Build NOT gate using IC 740
2. Build AND gate using IC 740
3. Build OR gate using IC 7432
4. Verify truth tables
5. Record observations







Deliverables

Students must submit:

1. Breadboard photo of each circuit
2. Completed truth tables
3. Short explanation of observed behavior

- Example:

Gate	Input(s)	Output
NOT	0	1
NOT	1	0